RAG systems landscape, late 2026: open source and managed

This note surveys the most advanced retrieval-augmented generation (RAG) systems available in late 2026: open-source projects and managed vendor offerings. For each it covers technique, what makes it advanced, maturity and licence, and where it fits best. It ends with a comparison table and three techniques worth adopting in this repo's grounded-extraction pipeline.

Status: **research note, 2026-10-31.** Every fact links to the primary page it was read on. Star counts, licences and versions are as shown on the GitHub page on **2026-10-31**. The GitHub API was not reachable from this session, so star counts are the rounded figures GitHub shows (for example "36.5k"), and most repo pages showed no release tag. Where a version is missing below, the page did not show one.

Evidence tags:

- **[verified]**: read on the linked primary page on 2026-10-31.
- **[vendor claim]**: a benchmark or quality claim the vendor or authors report about their own system. Not independently checked.
- **[ours]**: a judgement or recommendation made in this note.

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1. Summary

- **Graph RAG is now standard, and the lighter variants are the ones that get used.** Microsoft GraphRAG (community summaries, global/local/DRIFT search), LightRAG (dual-level graph+vector, 40.5k stars), HippoRAG 2 (Personalized PageRank) and Cognee (graph memory) all build an entity graph at index time. LightRAG has absorbed multimodal parsing (RAG-Anything merged in 2026.05).
- **Agentic retrieval is in every managed offering.** Azure AI Search knowledge bases, Bedrock Managed Knowledge Base and RAGFlow 1.0 all decompose a query into sub-queries, retrieve iteratively and check whether the results suffice. On the research side, Search-R1 trains this loop with reinforcement learning.
- **The retrieval core has settled.** Almost every system uses hybrid lexical+dense search fused with RRF, then a cross-encoder reranker. OpenAI exposes the RRF weights directly. Anthropic's Contextual Retrieval study puts numbers on each layer: 35% / 49% / 67% fewer retrieval failures [vendor claim].
- **Retrieval over page images is mature.** ColPali/ColQwen late interaction over page images (the `colpali-engine` package is now deprecated in favour of Sentence Transformers), Cohere Embed 5 (multimodal, 128k context) and Gemini File Search with `gemini-embedding-2` all index pages as images.
- **"Vectorless" reasoning retrieval is the notable newcomer.** PageIndex (39.9k stars) has an LLM walk a table-of-contents tree instead of using embeddings, and claims 98.7% on FinanceBench [vendor claim].
- **For this repo** [ours]: (1) Contextual Retrieval on top of the existing BM25+embedding hybrid,
    2. a structure-aware or tree-navigation step for long annual reports, PageIndex-style, (3) a second, model-independent groundedness check (HHEM-style NLI) beside `grounding.py`'s quote matching. Claude's native Citations API does not fit as-is, because it cannot be combined with structured outputs (§4.2).

2. Technique vocabulary

Technique	What it is	Where seen
Hybrid search + RRF	BM25/keyword and dense vectors run in parallel, merged by reciprocal rank fusion	R2R, OpenAI, Azure, Kotaemon
Cross-encoder rerank	A second model scores (query, chunk) pairs on a short list	Cohere Rerank 5, Contextual AI, LightRAG, Bedrock
Graph RAG	An LLM extracts entities and relations at index time; retrieval walks or summarises the graph	GraphRAG, LightRAG, HippoRAG, Cognee, Bedrock+Neptune
Late interaction	Token- or patch-level vectors scored with MaxSim	ColBERTv2/PLAID, ColPali/ColQwen
Agentic retrieval	An LLM plans sub-queries, retrieves in a loop and judges whether the results suffice	Azure, Bedrock Managed KB, RAGFlow, R2R, Search-R1
Contextual chunks	Each chunk gets an LLM-written summary of where it sits in the document before indexing	Anthropic Contextual Retrieval
Reasoning/tree retrieval	An LLM navigates a document's heading tree instead of using vector search	PageIndex, RAGFlow "Knowledge Compilation"
Groundedness scoring	An NLI-style model checks that the answer is supported by its evidence	Vectara HHEM, Contextual AI

3. Open-source projects

3.1 Microsoft GraphRAG

- **Technique.** The corpus is split into TextUnits. An LLM extracts entities, relationships and claims, and the graph is clustered hierarchically with Leiden. Community summaries are built bottom-up. There are four query modes: Global (over the community summaries), Local (fanning out from entities), DRIFT (Local plus community context) and Basic (top-k vectors). [verified] https://microsoft.github.io/graphrag/
- **Why it is advanced.** It answers "global" sense-making questions over a whole corpus, which top-k chunk retrieval cannot. [ours]
- **Maturity.** MIT, 36.5k stars, no release tag shown on the repo page (2026-10-31). https://github.com/microsoft/graphrag
- **Best fit.** Thematic questions across large static corpora. Indexing is expensive, because every chunk goes through the LLM. [ours]

3.2 LightRAG (HKUDS)

- **Technique.** A dual-layer graph+vector index with five query modes: local, global, hybrid, naive and mix (the default). A reranker has been the default for mix queries since 2025.08. RAG-Anything was merged in 2026.05 (multimodal parsing via MinerU and Docling), along with four chunking strategies and per-role LLM configuration. OpenSearch storage arrived in 2026.03. [verified] https://github.com/HKUDS/LightRAG

- **Maturity.** MIT, 40.5k stars (2026-10-31), no release tag shown.
- **Best fit.** Graph RAG at much lower indexing cost than GraphRAG, with incremental updates. [ours]

3.3 RAGFlow (InfiniFlow)

- **Technique.** DeepDoc layout analysis, OCR and table recognition (CPU inference in the 1.0 open-source build), template-based explainable chunking, and "Knowledge Compilation" (2026-08-19), which turns documents into wikis, graphs, trees, PageIndex, mind maps and timelines. Agentic RAG has four thinking modes (Low/Medium/High/Ultra). [verified] https://github.com/infiniflow/ragflow
- **Maturity.** Apache-2.0, 93.2k stars (2026-10-31). The news section lists 1.0.0-rc1 on 2026-09-29, and the Docker instructions reference v1.0.0-rc3.
- **Best fit.** A self-hosted, full-stack RAG product with a UI, for complex PDFs. [ours]

3.4 HippoRAG / HippoRAG 2 (OSU NLP)

- **Technique.** "RAG + Knowledge Graphs + Personalized PageRank": a knowledge graph of passages and entities, ranked by PPR from the entities in the query, which models associative, multi-hop memory. HippoRAG 2: arXiv 2502.14802. [verified] https://github.com/OSU-NLP-Group/HippoRAG · https://arxiv.org/abs/2502.14802
- **Maturity.** MIT, 4.1k stars (2026-10-31). A research codebase.
- **Best fit.** Multi-hop QA research and long-term memory. [ours]

3.5 R2R (SciPhi)

- **Technique.** Hybrid semantic+keyword search with RRF, automatic entity and relation graphs, an agentic reasoning agent, and a Deep Research API over the knowledge base and/or the web. [verified] https://github.com/SciPhi-AI/R2R
- **Maturity.** MIT, 8.1k stars (2026-10-31), no release tag shown.
- **Best fit.** A RAG backend you run behind a REST API. [ours]

3.6 Cognee

- **Technique.** Knowledge-graph memory: text becomes entities, relations and chunks, and code becomes a symbol and dependency graph. Retrieval picks graph, vector or code context. Lessons from a session can be distilled into permanent memory (`improve`). [verified] https://github.com/topoteretes/cognee
- **Maturity.** Apache-2.0, 32.9k stars, v1.6.1 (2026-09-24), as seen 2026-10-31.
- **Best fit.** Persistent agent memory. [ours]

3.7 LlamaIndex

- **Technique.** A framework: connectors, indices and graphs, Workflows, and an agent builder. The company's main focus is now LlamaParse (agentic OCR and parsing for 130+ formats, plus extraction). The README says the OSS framework "is no longer the company's primary focus". [verified] https://github.com/run-llama/llama_index
- **Maturity.** MIT, 52.8k stars (2026-10-31). This repo already uses it (`llama_index.retrievers.bm25` in `backend/arp/retrieval/index_cache.py`).
- **Best fit.** Gluing retrievers together, and parsing. [ours]

3.8 Haystack (deepset)

- **Technique.** Modular pipelines with loops and branches (sync, async and streaming), and agents with lifecycle hooks (`before_llm` , `before_tool` , `on_exit`) and concurrent tool calls. [verified] https://github.com/deepset-ai/haystack
- **Maturity.** Apache-2.0, 26.9k stars (2026-10-31).
- **Best fit.** Production pipelines that need explicit control flow. [ours]

3.9 DSPy (Stanford)

- **Technique.** A programming model that optimises prompts and weights for LM pipelines, RAG included. The repo lists the GEPA paper (reflective prompt evolution, July 2025). [verified] https://github.com/stanfordnlp/dspy
- **Maturity.** MIT, 38.9k stars (2026-10-31).
- **Best fit.** Tuning extraction and retrieval prompts against a labelled set, such as a golden set. [ours]

3.10 ColPali / ColQwen (visual retrieval)

- **Technique.** A vision-language model embeds page-image patches into multi-vectors, scored against query tokens with ColBERT-style late interaction. This skips the OCR and layout pipeline. Models: ColPali v1.1–1.3 (PaliGemma-3B), ColQwen2, ColQwen2.5 (Qwen2.5-VL-3B), ColSmol 256M/500M. Paper: arXiv 2407.01449. [verified] https://github.com/illuin-tech/colpali
- **Maturity.** MIT, 2.8k stars (2026-10-31). The README says `colpali-engine` “is now deprecated” and points new projects to Sentence Transformers.
- **Best fit.** Charts, tables and scanned pages, where text extraction loses structure. [ours]

3.11 Kotaemon (Cinnamon)

- **Technique.** A RAG UI with full-text+vector hybrid search, reranking, GraphRAG options (nano-GraphRAG, LightRAG, MS GraphRAG) and citations highlighted in an in-browser PDF viewer. [verified] https://github.com/Cinnamon/kotaemon
- **Maturity.** Apache-2.0, about 25.8k stars (2026-10-31).
- **Best fit.** Document QA with a human-checkable citation UX. [ours]

3.12 txtai (NeuML)

- **Technique.** An “embeddings database” that combines sparse and dense indexes, graph networks and relational storage. It adds LLM pipelines, workflows, smolagents-based agents and an MCP API. The page links “What’s new in txtai 9.0”. [verified] https://github.com/neuml/txtai
- **Maturity.** Apache-2.0, 13.1k stars (2026-10-31).
- **Best fit.** An all-in-one, lightweight semantic search engine in Python. [ours]

3.13 Agentic / deep-research RAG: Search-R1

- **Technique.** An RL framework (on veRL; PPO, GRPO, REINFORCE) that trains LLMs to interleave reasoning with search calls. Papers: arXiv 2503.09516, plus the follow-up study 2505.15117. [verified] https://github.com/PeterGriffinJin/Search-R1 · https://arxiv.org/abs/2503.09516
- **Maturity.** Apache-2.0, about 5.6k stars (2026-10-31). Research.
- **Best fit.** Training your own small search agent. Not relevant when Claude is used through the API. [ours]

3.14 Late interaction: ColBERTv2 / PLAID

- **Technique.** Token-level passage matrices scored with MaxSim. ColBERTv2 adds residual compression (arXiv 2112.01488), and PLAID is the efficient search engine for it (arXiv 2205.09707). [verified] https://github.com/stanford-futuredata/ColBERT
- **Maturity.** MIT, 3.9k stars (2026-10-31).
- **Best fit.** High-precision first-stage retrieval that is cheaper than a cross-encoder. [ours]

3.15 Newer: PageIndex (VectifyAI)

- **Technique.** “Reasoning-based RAG” with no vectors and no chunking: it builds a hierarchical tree index per document, and an LLM reasons down the tree to the relevant sections. It claims 98.7% on FinanceBench against about 50% for vector RAG [vendor claim]. [verified] https://github.com/VectifyAI/PageIndex
- **Maturity.** MIT, 39.9k stars (2026-10-31).
- **Best fit.** Long, well-structured financial and regulatory reports. This is the closest match to this repo’s inputs. [ours]

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4. Commercial / managed offerings

4.1 OpenAI file search (Responses API) and vector stores

- A hosted `file_search` tool over vector stores, with “semantic and keyword search”, `file_citation` annotations and attribute `filters`. [verified] https://developers.openai.com/api/docs/guides/tools-file-search
- Retrieval guide: RRF hybrid with `embedding_weight` / `text_weight`, a `ranker` and a `score_threshold`, and `rewrite_query`. Default chunks are 800 tokens with 400 overlap (allowed range 100–4096). Search returns up to 50 results. Storage is free for the first 1 GB, then $0.10/GB/day. [verified] https://developers.openai.com/api/docs/guides/retrieval
- **Best fit.** The quickest managed RAG for an OpenAI-model app. [ours]

4.2 Anthropic: Citations and Contextual Retrieval

- **Citations.** Claude returns `char_location` (plain text), `page_location` (PDF) or `content_block_location` (custom content) citations, and also cites `search_result` blocks. `cited_text` does not count toward output tokens. **Citations and structured outputs are incompatible:** combining them returns a 400 error. [verified] https://platform.claude.com/docs/en/build-with-claude/citations
- **Contextual Retrieval.** Claude writes a 50–100-token context for each chunk, which is prepended before embedding and before BM25 indexing. Measured failure rate for top-20 retrieval: 5.7% baseline, 3.7% with contextual embeddings (−35%), 2.9% adding contextual BM25 (−49%), and 1.9% adding a reranker (−67%; 150 candidates reranked down to 20). The one-time cost is about $1.02 per million document tokens with prompt caching [vendor claim]. [verified] https://www.anthropic.com/news/contextual-retrieval
- **Best fit.** Any Claude-based pipeline. Contextual Retrieval works regardless of which vector store you use. [ours]

4.3 Google: Vertex AI RAG Engine and Gemini File Search

- **RAG Engine.** Ingestion, chunking, embedding, corpus indexing, retrieval and generation. Vector backends: RagManagedDb, Vector Search 2.0, Feature Store, Weaviate, Pinecone and others. Parsing: the Document AI layout parser or an LLM parser. A reranking guide is linked. GA varies by region: GA in europe-west3/4, and allowlist or preview elsewhere. [verified] https://docs.cloud.google.com/vertex-ai/generative-ai/docs/rag-engine/rag-overview
- **Gemini File Search.** A managed semantic store with configurable chunking (`chunking_config`), citations carrying the file and page number, metadata filters, and multimodal indexing with `gemini-embedding-2` . Indexing is billed at embedding rates; storage and query embeddings are free. It cannot be combined with Google Search grounding in the same request. [verified] https://ai.google.dev/gemini-api/docs/file-search
- **Best fit.** GCP shops (RAG Engine), or the cheapest hosted RAG for Gemini apps (File Search). [ours]

4.4 AWS Bedrock Knowledge Bases

- There are two kinds. **Managed** KB: AWS-managed embedding and reranking, Smart Parsing per document type (PDF, PPTX, scans, audio, video), agentic retrieval (multi-hop, sub-queries, sufficiency checks), ACL filtering, and MCP via AgentCore Gateway. **Customer-managed** KB: your own vector store (OpenSearch Serverless, Aurora, Neptune). Other options include Neptune Analytics graphs (GraphRAG), structured data stores, Kendra GenAI indexes, multimodal image queries, rerankers and citations. [verified] https://docs.aws.amazon.com/bedrock/latest/userguide/knowledge-base.html
- **Best fit.** AWS enterprises that need connectors and ACLs. [ours]

4.5 Azure AI Search agentic retrieval (Foundry IQ)

- A knowledge base plans sub-queries with an LLM (at `low` / `medium` reasoning effort), runs them in parallel as keyword, vector or hybrid search, semantically reranks each one, and merges the results with references and an activity log. GA REST API `2026-04-01` covers minimal effort; `2026-08-01-preview` adds query planning, answer synthesis and multi-turn. Billing is token-based. [verified] https://learn.microsoft.com/en-us/azure/search/agentic-retrieval-overview
- **Best fit.** Microsoft and Foundry agent stacks. [ours]

4.6 Cohere Rerank 5 and Embed 5

- **Rerank 5:** `rerank-v5.0-pro` and `-fast` , one multilingual model (100+ languages), and YAML for structured documents. [verified] https://docs.cohere.com/docs/rerank-overview
- **Embed 5:** `embed-v5.0-pro` and `-fast` in a shared space, with text, image and mixed input (PDF pages), 128k context, Matryoshka dimensions from 256 to 2048, and float, int8 or binary output. [verified] https://docs.cohere.com/docs/cohere-embed
- **Best fit.** A managed reranker or embedder you can drop into any pipeline, including German text. [ours]

4.7 Vectara

- An “API-first Agentic Platform” with multimodal retrieval, hallucination evaluation and correction. [verified] https://docs.vectara.com/docs/
- HHEM: HHEM-2.1-Open (Apache-2.0, no context limit) scores whether a premise supports a hypothesis on a 0–1 scale. HHEM-2.3 is commercial. [verified]

https://huggingface.co/vectara/hallucination_evaluation_model
- **Best fit.** Regulated RAG that needs a groundedness score. [ours]

4.8 Contextual AI

- The docs list a parser, an instruction-following reranker, a grounded language model, groundedness scoring and agents. No reranker version was visible on that page. [verified] https://docs.contextual.ai/
- **Best fit.** Enterprise RAG where groundedness is the selling point. [ours]

4.9 Glean

- The vendor describes an Enterprise Graph (people, content, activity, synced permissions), an agentic engine that chooses between fast search and a multi-step plan, permissions enforced before the LLM, and semantic+keyword+graph ranking with cited answers [vendor claim]. https://www.glean.com/blog/agentic-reasoning-future-ai · https://www.glean.com/blog/from-enterprise-search-to-enterprise-context-what-ai-agents-actually-need
- **Best fit.** Workplace search across SaaS tools, not document extraction. [ours]

4.10 Pinecone Assistant

- Managed chunking, embedding and storage, answers with file references, a Context API for your own LLM, and an MCP server. [verified] https://docs.pinecone.io/guides/assistant/overview
- **Best fit.** Teams already on Pinecone. [ours]

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5. Comparison table

System	Graph	Hybrid	Rerank	Agentic loop	Multimodal	Licence / model	Stars (2026-10-31)
MS GraphRAG	✓ (Leiden communities)	–	–	DRIFT	–	MIT	36.5k
LightRAG	✓	✓ (mix)	✓ default	–	✓ (RAG-Anything)	MIT	40.5k
RAGFlow	✓ (compilation)	✓	✓	✓ 4 modes	✓ DeepDoc	Apache-2.0	93.2k
HippoRAG 2	✓ PPR	–	–	–	–	MIT	4.1k
R2R	✓	✓ RRF	–	✓ deep research	–	MIT	8.1k
Cognee	✓ memory	✓	–	–	–	Apache-2.0	32.9k
LlamaIndex	✓	✓	✓	✓ workflows	via LlamaParse	MIT	52.8k
Haystack	–	✓	✓	✓ agents	–	Apache-2.0	26.9k
DSPy	–	–	–	optimiser	–	MIT	38.9k
ColPali/ColQwen	–	–	late interaction	–	✓ page images	MIT	2.8k
Kotaemon	✓	✓	✓	–	PDF highlight	Apache-2.0	25.8k

System	Graph	Hybrid	Rerank	Agentic loop	Multimodal	Licence / model	Stars (2026-10-31)
txtai	✓	✓ sparse+dense	–	✓	✓	Apache-2.0	13.1k
Search-R1	–	–	–	✓ RL-trained	–	Apache-2.0	5.6k
ColBERT/PLAID	–	–	late interaction	–	–	MIT	3.9k
PageIndex	tree	no vectors	–	✓ LLM tree walk	–	MIT	39.9k
OpenAI file search	–	✓ RRF weights	✓ ranker	model-driven	–	managed, $0.10/GB/day >1 GB	–
Anthropic	–	contextual BM25+emb	recommended	–	PDF citations	API feature	–
Vertex RAG Engine / Gemini FS	–	– / semantic	✓	–	✓ gemini-embedding-2	managed	–
Bedrock KB	✓ Neptune	✓	✓	✓ Managed KB	✓ Smart Parsing	managed	–
Azure agentic retrieval	–	✓	✓ semantic	✓ query planning	–	managed, token billing	–
Cohere Rerank 5 / Embed 5	–	–	✓	–	✓ Embed 5	managed	–
Vectara	–	✓	✓	✓	✓	managed; HHEM-2.1 open	–
Contextual AI	–	–	✓ instruction-following	✓	parser	managed	–
Glean	✓ enterprise graph	✓	✓	✓	–	managed	–
Pinecone Assistant	–	–	–	–	–	managed	–

"–" means not stated on the page read, not proven absent.

6. Relevance to this repo

What exists now (skimmed 2026-10-31):

- Evidence retrieval is BM25 by default (`llama_index` BM25Retriever, `backend/arp/retrieval/index_cache.py`), with an optional hybrid mode using BM25 plus local multilingual embeddings (`Settings.hybrid_retrieval_enabled`, `backend/arp/extraction/steps.py`) and an optional local cross-encoder rerank that is off by default because of CPU cost (`backend/arp/retrieval/rerank.py`). German keyword expansion lives in `backend/arp/retrieval/keywords_de.py`.
- Grounding is post-hoc: `backend/arp/grounding.py` normalises text, finds each quoted citation in the source by exact or fuzzy match (threshold 0.92, numbers need a label below 20 characters),

and maps the match to a page, sheet and table. `backend/arp/extraction/vision_check.py` checks values against page images.

Recommendations [ours]:

1. **Contextual Retrieval on the existing hybrid index.** Prepend a Claude-written, prompt-cached context line to each chunk before both BM25 and embedding indexing. It fits the current architecture directly (both indexes already exist), and it is the only technique here with a published ablation on the same BM25+embedding+rerank stack: −49% failures without a reranker, which matters because the rerank step is off by default. Corporate reports suffer from the problem it fixes: a chunk saying "Scope 1 fell 12%" without naming the year or entity. Keep the context out of `grounding.py`'s source text so quotes are still matched against the original document.
2. **Structure-aware retrieval for long reports (PageIndex-style tree navigation).** Annual and sustainability reports have reliable tables of contents and section headings. Have Claude pick sections from the heading tree first, then run BM25 inside them. This cuts recall misses where the keywords are unusual. Treat PageIndex's FinanceBench number as unverified and measure on the golden set before adopting it.
3. **A second, NLI-style groundedness check** (HHEM-2.1-Open, Apache-2.0, or a Claude judge) after `grounding.py`. Quote matching proves a quote *exists*. It does not prove the quote *supports the extracted value* (unit, year, scope). A premise/hypothesis score on (quote, field+value) would feed the existing routing rules (`backend/arp/extraction/routing.py`) as one more quality signal.

Considered and not recommended now:

- **Claude native Citations.** It would give exact char and page spans for free, but it cannot be used together with structured outputs (400 error). Adopting it would mean dropping JSON-schema output on the extraction calls. `grounding.py` already gives the same assurance after the fact.
- **Full graph RAG** (GraphRAG/LightRAG). Extraction asks field-level questions, not global sense-making ones, and the indexing cost does not pay off. Revisit for cross-company questions.
- **ColQwen visual retrieval.** Worth a trial only if the golden set shows misses on chart- or table-only pages. The vision check already covers verification.

`openwiki/` needs no update: this note changes no behaviour.